

# Prateek Kumar Sahu

PhD Candidate

Department of Psychology

University of Alberta, CA

Proficiency in coding languages: **R** (Expert), Python (Moderate)

Proficient in Causal inference, **Bayesian** modeling, **Bio-acoustics**

## Contact

psahu@ualberta.ca

<https://prateeksahu.com>

[LinkedIn](#), [GitHub](#)

## Education

- University of Alberta** Edmonton, Canada  
*PhD, Department of Psychology* 2022 - Now
- University of Alberta** Edmonton, Canada  
*MSc, Department of Psychology* 2019 - 2022
- Indian Institute of Science Education and Research** Kolkata, India  
*BSc & MSc, Department of Biological Sciences* 2013 - 2018

## Research Experience

- Animal cognition, Vocal production and perception, Novel bioacoustic techniques, open-source research hardware design, preferences in songbirds Doctoral project  
*Dr. Christopher Sturdy* 2022-now
- Methodological comparisons of classification techniques and refinement of an operant go/no-go procedure Masters' thesis  
*Dr. Christopher Sturdy* 2019-2022
- Neural basis of sugar-elicited search behaviour of *Drosophila* Research Project  
*Dr. Axel Brockmann* 2018-2019
- Understanding reunification dynamics in ant colonies Summer Project, Masters' thesis  
*Dr. Sumana Annagiri* 2015, 2017-2018
- Consistency in personality traits of the shoals of climbing perch Summer Project  
*Dr. V. V. Binoy* 2016

## Publications

- Sahu, P.K.**, Nottebrock, K., Ratch, J., Smeltz, S.M.L., Stenstrom, K.H., Deimeke, M.J., & Sturdy, C.B. (2025) Differential response variability of black-capped chickadees to wingbeat sounds and vocalizations. *Biology Open*, 15(6), bio062598, <https://doi.org/10.1242/bio.062598>
- Stenstrom, K.H., Deimeke, M.J., **Sahu, P.K.**, Smeltz, S.M.L., & Sturdy, C.B. (2026). What is rescue anyways: A comprehensive review of rescue and rescue adjacent behavior across taxa. *Animal Behavior and Cognition*, 13(2), 197-213. <https://doi.org/10.26451/abc.13.02.06.2026>

11. *Stenstrom, K.H., Deimeke, M.J., Sahu, P.K., Smeltz, S.M.L., & Sturdy, C.B.* (2026). Risk versus reward: A systematic review on the threat mobbers face. *Animal Behavior and Cognition*, 13(2), 184-196. <https://doi.org/10.26451/abc.13.02.05.2026>
10. *Sahu, P.K., Deimeke, M.J., Garcia, A.D., Stenstrom, K.H., Smeltz, S.M.L., Wang, Y., Ratch, J., & Sturdy, C.B.* (2026) Re-examining hidden fitness: Female preferences for long-path songs in zebra finches. *PLOS ONE*, 21(3):e0343886, <https://doi.org/10.1371/journal.pone.0343886>
9. *Sahu, P.K., Garcia, A.D., Stenstrom, K.H., Deimeke, M.J., Smeltz, S.M.L., Lank, I.J., & Sturdy, C.B.* (2025) A cost-effective Raspberry pi-based operant playback setup to evaluate auditory preferences in songbirds. *Behavioural Processes*, 232, 105276, <https://doi.org/10.1016/j.beproc.2025.105276>
8. *Smeltz, S.M.L., Deimeke, M.J., Montenegro, C., Sahu, P.K., Stenstrom, K.H., Camacho-Alpizar, A., & Sturdy, C.B.* (2025) Individual discrimination within, but not between, two vocalization types of the black-capped chickadee. *Ethology*, e13572. <https://doi.org/10.1111/eth.13572>
7. *Smeltz, S.M.L., Deimeke, M.J., Sahu, P.K., Montenegro, C., Stenstrom, K.H., Starenchak, I., Rennie, V., Ebrahim, I.M., Jones, J.A., & Sturdy, C.B.* (2025) Individual vocal identity may be obscured following colony assembly in captive black-capped chickadees. *Behavioural Processes*, 225, 105151, <https://doi.org/10.1016/j.beproc.2025.105151>
6. *Sahu, P.K., Montenegro, C., Lambert, C.T., Oprea, A., Deimeke, M.J., Rennie, V., Smeltz, S.M.L., Benowicz, T.J., Patel, D., Phillmore, L.S., & Sturdy, C.B.* (2022) Effect of feed-time duration on discrimination of vocalizations in a go/no-go operant paradigm. *Behavioural Processes*, 203, 10477, <https://doi.org/10.1016/j.beproc.2022.104777>
5. *Sahu, P.K., Campbell, K.A., Oprea, A., Phillmore, L.S., & Sturdy, C.B.* (2022) Comparing methodologies for classification of zebra finch distance calls. *The Journal of the Acoustical Society of America*, 151, 3305, <https://doi.org/10.1121/10.0011401>
4. *Lambert, C.T., Sahu, P.K., Sturdy, C.B., & Guillette, L.M.* (2022) Among-individual differences in auditory and physical cognitive abilities in zebra finches. *Learning & Behaviour*, 50(3), 389-404, <https://doi.org/10.3758/s13420-022-00520-w>
3. *Montenegro, C., Sahu, P.K., & Sturdy, C.B.* (2021). Individual differences in female black-capped chickadee (*Poecile atricapillus*) fee-bee songs. *The Journal of the Acoustical Society of America*, 150(4), 3038-3046, <https://doi.org/10.1121/10.0006532>
2. *Montenegro, C., Service, W.D., Scully, E.N., Mischler, S.K., Sahu, P.K., Benowicz, T.J., Fox, K.V. R., & Sturdy, C.B.* (2021). The impact of anthropogenic noise on individual identification via female song in Black-capped chickadees (*Poecile atricapillus*). *Scientific Reports*, 11(1), <https://doi.org/10.1038/s41598-021-96504-3>
1. *Sahu, P.K., Kolay, S. & Annagiri, S.* (2019) To reunite or not: A study of artificially fragmented *Diacamma indicum* ant colonies. *Behavioural Processes* 158, 4-10, <https://doi.org/10.1016/j.beproc.2018.10.017>

## Teaching Experience & Service

- Teaching Assistant: Basic Psychological Processes (2019, 2020, 2021, 2022), Behaviour Modification (2020, 2021, 2022, 2023, 2024, 2025), Principle of Learning (2020), Behaviour & Genetics (2021), Neuroethology (2022), Honours Seminar: Writing course (2023), Theory in Learning and Comparative Cognition (2024), Social Influence (2024, 2025), Research Methods Cognition (2025), Human Neuropsychology (2025)
- Skype A Scientist: Talks at various schools (Indonesia, New York, Colorado, New York) (2022-2025)
- Secretary/Treasurer of the Graduate Psychology Association: (2023-2026)
- Judge for the Royce Research Conference, University of Alberta, CA: (2021, 2024, 2025)
- Volunteer for organizing the International Conference on Comparative Cognition, USA: (2023, 2025)

## Awards & Scholarships

|                                                                                |         |
|--------------------------------------------------------------------------------|---------|
| Alberta Graduate Excellence Scholarship (AGES), CA . . . . .                   | 2025-26 |
| Graduate Student Travel Award, CA . . . . .                                    | 2024-25 |
| Comparative Cognition Society- Ron Weisman Travel Award, USA . . . . .         | 2025    |
| Alberta Graduate Excellence Scholarship (AGES), CA . . . . .                   | 2024-25 |
| Alberta Graduate Excellence Scholarship (AGES), CA . . . . .                   | 2023-24 |
| Scholarship from Department of Psychology, University of Alberta, CA . . . . . | 2019-22 |
| INSPIRE Fellowship, IN . . . . .                                               | 2013-18 |
| National Means cum Merit Scholarship, IN . . . . .                             | 2008-10 |

## Conferences

- |                                                                            |      |
|----------------------------------------------------------------------------|------|
| • Royce Research Conference                                                | 2026 |
| • International Conference On Comparative Cognition                        | 2026 |
| • Royce Research Conference                                                | 2025 |
| • International Conference On Comparative Cognition                        | 2025 |
| • The Canadian Society for Brain, Behaviour and Cognitive Science (CSBBCS) | 2024 |
| • International Conference On Comparative Cognition                        | 2023 |
| • International Conference On Comparative Cognition                        | 2021 |
| • Animal Behaviour Society Conference                                      | 2020 |
| • Virtual Royce Conference                                                 | 2020 |
| • Insect Multisensory Integration Workshop                                 | 2019 |
| • 2nd Bangalore Meeting on Asian Bees                                      | 2019 |

- Preparatory School on Population Genetics and Evolution 2019
- Understanding Behaviour 2019
- Workshop on Cyberspace and Cyber Security 2017
- The Ambassadors of Evolution Workshop 2017
- The 10th Year Celebration of IISER Kolkata Advances in Life Sciences 2017
- The Frontiers in Modern Biology (FIMB) symposium 2015

Manuscript Reviews

1.*Ostrich (2022)*; 2,3.*PeerJ (2022, 2023)*, 4.*Behavioral Ecology and Sociobiology (2022)*, 5.*Animal Cognition (2023)*, 6.*Animal Behaviour (2023)*, 7.*Animal Behaviour (2023)*, 8.*The Journal of the Acoustical Society of America (2023)*, 9.*Journal of Comparative Psychology (2023)*, 10.*PLOS One (2024)*, 11.*IBIS: International Journal of Avian Science (2024)*, 12.*Journal of Experimental Biology (2025)*, 13,14.*Behavioural Processes (2025, 2025)*, 16,17) *Learning and Motivation (2026, 2026)*

Skills

Bio-acoustical techniques & animal learning paradigms . . . . . University of Alberta, 2019-now  
Handling birds & animal care . . . . . University of Alberta, 2019-now  
Raising & handling Drosophila . . . . . National Centre for Biological Sciences, 2018-19  
Insect(ants, bees) research & insect biology . . . . . IISER-K,NCBS 2015-19  
Fish research & handling . . . . . National Institute of Advanced Studies, 2016  
Coding & Statistics . . . . . "R","Python", "Matlab"  
Graphics Designing . . . . . "Photoshop", "Illustrator"